



Depth-induced prompt learning for laparoscopic liver landmark detection

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ABSTRACT

Laparoscopic liver surgery presents a highly intricate intraoperative environment with significant liver deformation, posing challenges for surgeons in locating critical liver structures. Anatomical liver landmarks can greatly assist surgeons in spatial perception in laparoscopic scenarios and facilitate preoperative-to-intraoperative registration. To advance research in liver landmark detection, we develop a new dataset called *L3D-2K*, comprising 2000 keyframes with expert landmark annotations from surgical videos of 47 patients. Accordingly, we propose a baseline, *D²GPLand+*, which effectively leverages depth modality to boost landmark detection performance. Concretely, we introduce a Depth-aware Prompt Embedding (DPE) scheme, which dynamically extracts class-related global geometric cues with the guidance of self-supervised prompts from the SAM encoder. Further, a Cross-dimension Unified Mamba (CUMamba) block is designed to comprehensively incorporate RGB and depth features with the concurrent spatial and channel scanning mechanism. Besides, we bring out an Anatomical Feature Augmentation (AFA) module that captures anatomical cues and emphasizes key structures by optimizing feature granularity. For benchmarking purposes, we evaluate our method and 17 mainstream detection models on *L3D*, *L3D-2K*, and *P2ILF* datasets. Experimental results demonstrate that *D²GPLand+* obtains superior performance on all three datasets. Our approach provides surgeons with guiding clues that facilitate surgical operations and decision-making in complex laparoscopic surgery. Our code and dataset are available at <https://github.com/cuiruize/D2GPLand-Plus>.

1. Introduction

Driven by the pursuit of minimally invasive treatments and excellent clinical outcomes, laparoscopic liver surgery has attracted attention thanks to its advantages of reduced trauma (Schneider et al., 2021; Guo et al., 2025). However, distinguishing critical anatomical structures in liver surgeries remains a challenging problem due to significant intraoperative liver deformation and intricate laparoscopic environment, which relies heavily on the expertise and experience of the surgeon. In this regard, there is an urgent need for augmented reality (AR) navigation techniques in laparoscopic liver surgery to provide surgeons with auxiliary information for precise resection and early warning of surgical risks. The primary step toward AR-assisted intervention is automatically identifying guiding markers that can reliably locate liver from keyframes of intraoperative 2D videos and align with the consistent preoperative

3D (Zhou et al., 2025). Liver anatomical landmarks, e.g., anterior ridge and falciform ligament, have been validated as effective biomarkers for 2D-3D fusion (Labrunie et al., 2022, 2023). As shown in the top of Fig. 1, with 2D and 3D landmarks as references, preoperative liver anatomy can be projected onto intraoperative keyframes to showcase a comprehensive surgical scene for enhanced visual guidance. Specifically, liver landmarks establish robust correspondences and alignments of regions of interest (RoI) between preoperative 3D anatomy and intraoperative 2D laparoscopic keyframes, hence guiding surgeons for precise surgical actions and assisting in decision-making. Moreover, given the limited surface texture features of the liver, liver landmarks can also serve as anatomical reference markers for camera-pose estimation in 2D-3D liver registration (Zhang et al., 2025a; Han et al., 2025).

However, the lack of annotated datasets and the insufficient exploitation of geometric information in video frames have hindered progress in

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accurate and automatic laparoscopic landmark detection research. Liver landmarks in laparoscopic keyframes are typically represented as points, contours, or bounding boxes to provide location information (Collins et al., 2020; Smithmaitrie et al., 2024). Nonetheless, these conventional representations fail to provide semantic information to establish precise correspondence between intraoperative 2D keyframes and preoperative 3D liver anatomy, which is crucial for estimating cross-dimensional spatial relationships during laparoscopic liver surgery. To address these challenges, we adopt silhouette, ridge, and ligament as landmarks, which are also continuous anatomies with well-defined semantic attributes in preoperative 3D anatomy, and annotate them in the form of continuous semantic regions. Subsequently, we construct the largest currently available dataset of laparoscopic liver landmarks, named **L3D-2K**, to bridge the gap in annotated data for training deep learning-based landmark detection models (Ali et al., 2025; Cui et al., 2025). Specifically, we invite four senior surgeons to label 2000 keyframes selected from surgical videos of 47 patients from two medical sites, with three types of semantic landmarks annotated. Based on the L3D-2K benchmark, we contribute a systematic evaluation of 13 mainstream approaches (Ronneberger et al., 2015; Labrunie et al., 2023; Xiao et al., 2018; Chen et al., 2018; Zhou et al., 2019; Wang et al., 2020; Chen et al., 2021; Cao et al., 2022; Chen et al., 2023; Zhang and Liu, 2023; Hu et al., 2023; Chai et al., 2024; Pei et al., 2024), aiming to advance research in automatic liver landmark detection.

Furthermore, in intricate laparoscopic environments, the performance of existing methods is hindered by unstable detection precision owing to interruptions, tissue deformations, and the absence of global geometric information (Özgür et al., 2018). Current approaches (Koo et al., 2022; Labrunie et al., 2023, 2022) predominantly focus on extracting local semantic features and edge clues while ignoring global geometric characteristics of the liver region, especially depth information, which includes essential shape features for liver localization. In this regard, we delve into an effective framework that leverages depth modality and pre-trained large vision models (Kirillov et al., 2023) to improve the accuracy of laparoscopic liver landmark detection. Achieving this objective necessitates overcoming three key challenges: 1) the high similarity among landmark categories, which complicates precise feature extraction; 2) the difficulty of effectively integrating RGB and depth information; and 3) the inherent difficulty of learning robust anatomical representations in the presence of outliers (e.g., blood, smoke, and reflections) and ambiguous liver boundaries.

In this work, we introduce a depth-induced prompt learning network called **D²GPLand+** for laparoscopic liver landmark detection. By employing an off-the-shelf depth estimation model, we first generate depth maps that provide inherent anatomical information. Considering that Segment Anything Model (SAM)-based approaches (Kirillov et al., 2023; Wu et al., 2023) have shown superior performance in extracting global high-level features in surgical scenes, we adopt a pre-trained SAM encoder combined with a CNN encoder to respectively extract depth geometric information and RGB features. To distinguish highly similar landmark characteristics in laparoscopic liver surgery, we propose a *Depth-aware Prompt Embedding (DPE)* operation to highlight geometric attributes through prompt contrastive learning, resulting in class-aware geometric features. Building on the capabilities of Mamba (Gu and Dao, 2023; Zhu et al., 2024; He et al., 2025; Pei et al., 2025) to utilize state variables for sequential state modeling and achieve linear computational complexity through the selective scanning mechanism, we design a *Cross-dimension Unified Mamba (CUMamba)* block. It facilitates sufficient cross-modal feature interactions across spatial and channel dimensions to integrate RGB and depth features. Moreover, drawn from the principle that the human visual system adjusts edge granularity to distinguish objects, we introduce *anatomical Feature Augmentation (AFA)* module to enhance the learning of anatomical structures. Concretely, we utilize the Discrete Fourier Transform (DFT) to implicitly emphasize edges by capturing high-frequency components combined with a Laplacian anatomical filter to explicitly refine landmark cues. Addi-

tionally, we embed a learnable granularity variable to control the detail levels of these features. Extensive experimental results on L3D, L3D-2K, and P2ILF datasets demonstrate that **D²GPLand+** achieves a superior performance in the liver landmark detection task.

This work is an extension of our conference version (Pei et al., 2024) in terms of several aspects. First, we extend the scale of the dataset from 1152 to 2000 keyframes. Notably, we include more practical but challenging scenarios, which increases the authenticity and detection difficulty of liver landmarks. A comprehensive statistical analysis of the extended dataset is conducted. Second, we further upgrade our framework by introducing the CUMamba block to enhance the cross-modal interaction. Meanwhile, we replace the reverse attention with the AFA scheme to empower the learning of key landmark structures. Third, we compare **D²GPLand+** on the extended L3D-2K dataset and P2ILF dataset with cutting-edge methods to prove the effectiveness of our method.

Our contributions can be summarized as follows:

- We create a new liver landmark detection dataset, L3D-2K, comprising 2000 keyframes sourced from surgical videos with professional annotations.
- We present a novel framework **D²GPLand+** for laparoscopic liver landmark detection. Our model exploits depth information to interact with RGB features to perceive anatomical clues and accurately detect key landmarks.
- We design a DPE module that incorporates learnable prompts with contrastive learning to discriminate the geometric features of different landmark categories. We propose CUMamba that concurrently conducts cross-modal information exchange on spatial dimension and feature reparameterization on channel dimension for effective RGB-D fusion. The AFA scheme is introduced to highlight anatomical structures through implicit and explicit edge emphasis and controlling detail levels.
- Experimental results on L3D, L3D-2K, and P2ILF datasets demonstrate that **D²GPLand+** achieves a promising performance and outperforms 13 well-known methods for liver landmark detection.

2. Related works

2.1. Liver landmark detection methods

Most laparoscopic liver registration procedures involve the assistance of manually marked anatomical landmarks. Considering the time-consuming and laborious manual labeling, automatic landmark detection is essential for efficient intraoperative liver registration. According to the traditional definition that represents liver landmarks as contours, the straightforward approach is to employ the edge-detection network (e.g., CASENet Yu et al., 2017) to detect liver contours (Koo et al., 2022). Several methods (Le Roy et al., 2019; Chen et al., 2025) employ a CNN-based architecture to segment livers and depict the anatomical landmarks. Another popular strategy is to define liver landmarks as regions and consider landmark detection as a semantic segmentation task, which provides sufficient semantics and mitigates the effect of bias in model inferences (Schneider et al., 2021). The classical U-Net (Ronneberger et al., 2015) has been widely used for automatic landmark detection (Labrunie et al., 2022). More recently, advanced networks (Ali et al., 2025) like ResUnet (Xiao et al., 2018), and UNet++ (Zhou et al., 2019) have been applied for precise landmark detection. Similarly, another work (Zhang et al., 2025b) proposes a dual-decoder network that utilizes additional annotations of the liver surface and prior knowledge from the SAM encoder to enhance detection accuracy in AR-guided surgery scenarios. Following previous works (Labrunie et al., 2022; Ali et al., 2025), we define the liver landmark detection as a semantic segmentation task. Recognizing the importance of geometric information for identifying liver anatomical landmarks, this study introduces a novel strategy leveraging depth features for improved intraoperative liver landmark detection.

2.2. Datasets for liver landmark detection

Most laparoscopic liver landmark data are contained in liver registration datasets (Plantevefe et al., 2014; Adagolodjo et al., 2017; Koo et al., 2017), and few datasets are developed specifically for this task. Koo et al. (2022) construct an unpublished dataset for preoperative liver registration in laparoscopic liver resection surgery, which includes 1415 images with corresponding landmark annotations, i.e., ridge, ligament, and silhouette. These frames are extracted from 68 patients, with each patient contributing 10 to 30 frames at 1920×1080 resolution. Another unpublished dataset is proposed by Kasai et al. (2023), which includes only 380 frames extracted from laparoscopic videos with liver landmark annotations (silhouette and liver boundary). There is also a recent challenge dataset P2ILF (Ali et al., 2025) that consists of 183 laparoscopic images with annotations (ridge, ligament, and silhouette). This dataset splits 167 images for training while the remaining 16 images are used for testing. In this work, we aim to establish a comprehensive dataset that provides abundant, diverse, and high-quality samples for laparoscopic liver landmark detection.

2.3. RGB-D modal fusion

Depth information has been shown to be effective as auxiliary guidance in vision tasks. For semantic segmentation, RDFNet (Park et al., 2017) fuses RGB and depth features via a dual-branch structure, while PCGNet (Liu et al., 2022) enhances multi-scale RGB and depth modal fusion using a pyramid structure. For object detection tasks, depth features are widely exploited to align 2D-3D information (Tziafas and Kasaei, 2023). In the medical domain, combining depth and RGB modalities offers significant advantages. For instance, CS²-Net (Mou et al., 2021) proposes a 3D attention mechanism to aggregate depth into RGB features for precise segmentation of curvilinear structures. In laparoscopic keyframe analysis, depth data plays a critical role in recognizing liver landmarks, which deserves more attention in our task. Recently, Depth Anything Model (DAM) (Yang et al., 2024) yields remarkable performance in estimating depth maps for medical images (Han et al., 2024). In our framework, we adopt DAM to explore depth cues and blend with RGB data to improve the accuracy of landmark detection.

3. L3D-2K dataset

P2ILF (Ali et al., 2025) is the first publicly available dataset for intraoperative laparoscopic liver landmark detection. However, P2ILF only includes 183 laparoscopic images from 11 patients, which is insufficient for training deep learning-based models. In this regard, we established a new dataset, termed L3D-2K, to facilitate liver landmark detection in laparoscopic scenarios. In this section, we will describe the process of data collection and the statistical distribution of our dataset in detail.

3.1. Dataset construction

3.1.1. Keyframe collection

First, we invite four professional surgeons to select keyframes based on the following criteria:

- The keyframes should be at the time points where the surgeon needs to locate liver landmarks and operate with the guidance of preoperative-intraoperative registration.
- The keyframes should allow the surgeon to observe the global view of the liver, which can greatly reduce anatomical misperception during the surgery.
- The keyframes are supposed to cover as many types of liver landmarks as possible to achieve universal coverage of landmark information in the dataset.

We finally selected 2000 keyframes from laparoscopic liver surgical videos of 47 patients with Hepatocellular Carcinoma (HCC) in two part-

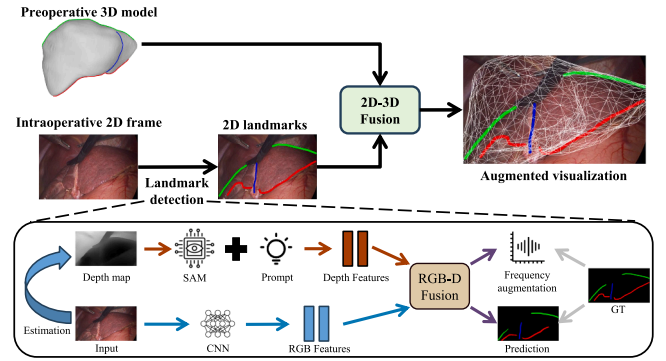


Fig. 1. Top: The motivation of detecting laparoscopic liver landmarks. Liver landmark detection is essential for preoperative-to-intraoperative registration to augment visual guidance. Bottom: Illustration of the proposed D²GPLand+. D²GPLand+ can exploit depth and RGB features to holistically perceive anatomical clues and precisely detect liver landmarks.

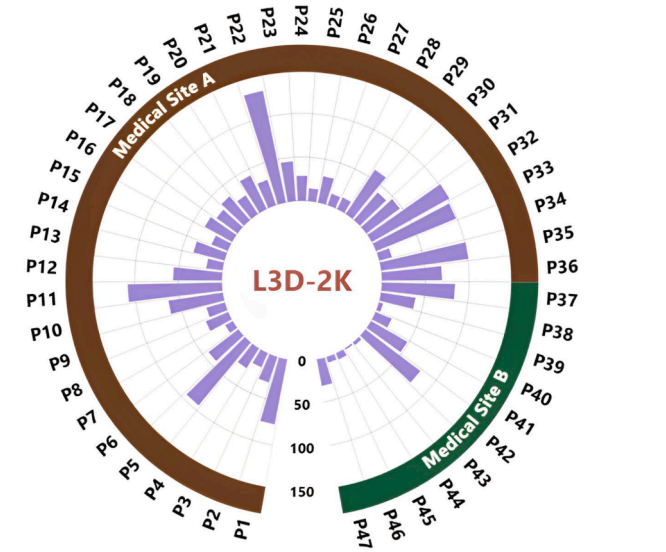


Fig. 2. Distribution of keyframe sources in L3D-2K.

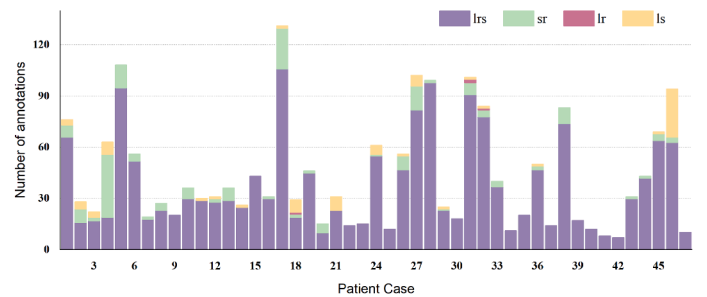


Fig. 3. Distribution of the landmark annotations. l, r, s are ligament, ridge, and silhouette, respectively.

ner hospitals. According to the suggestions from our collaborative surgeons, we select more practical but challenging samples to improve the authenticity and professionalism of our dataset.

To the best of our knowledge, L3D-2K is currently the largest publicly available intraoperative laparoscopic liver landmark detection dataset. Since the liver characteristics of the same patient are highly similar, randomly splitting all frame models may learn the data distribution in the training phase from the test or validation set. To ensure fairness, keyframes from the same patient are not shared across different sets.

Table 1
Attribute descriptions of patient cases in L3D-2K.

Attributes	Descriptions
Male, Female	The gender of the patient.
LL,RL,BL	Lobe locations. The liver tumor is located on the left, right, or on both liver lobes of the patient.
10-19,20-29,30-39,40-49,50-59,70-79,100+	The liver tumor size of the patient. The unit of tumor size is mm.
SF, MF	Number of Focuses. There is single focus or multiple focuses on the patient's liver.
CI	Cirrhosis. The patient has cirrhosis.

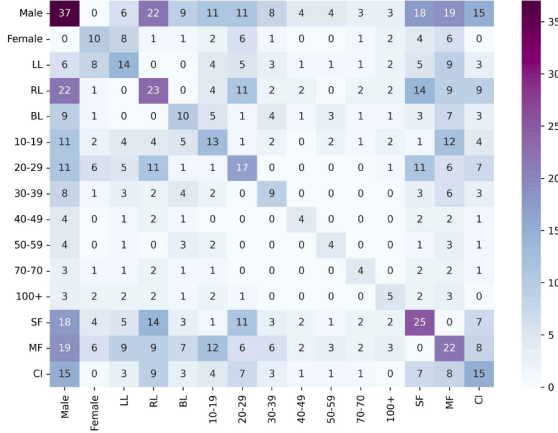


Fig. 4. Co-attribute distribution on L3D-2K. The number in each grid denotes the number of cases that fulfill attributes.

We divide all samples into three sets, where 1532 keyframes are used as the training set, 230 as the validation set, and 238 as the test set.

3.1.2. Landmark annotation

To ensure the authenticity of L3D-2K, the surgeons who selected the keyframes are invited to annotate landmarks. The surgeons label three categories of liver landmarks: silhouette, ridge, and ligament. Notably, landmarks are all annotated in the form of semantic regions to cover complete areas of critical anatomical structures. The labeling principles are as follows:

- The ridge landmark is defined as the lower anterior ridge of the liver.
- The ligament landmark is defined as the junction between two lobes of the liver.
- The visible silhouette landmark is defined as the upper anterior ridge of the liver.
- The landmarks that are obstructed by surgical instruments do not require annotation.

3.2. Dataset statistics and analysis

L3D-2K embraces keyframes from various patient cohorts from two medical sites to enhance data diversity. As shown in Fig. 2, we summarize the distribution of keyframe sources in the proposed dataset. 1,672 keyframes are provided by 36 patients from Medical Site A, while the other 328 keyframes are provided by 11 patients from Medical Site B. The number of keyframes provided by each patient varies from 2 to 261.

We also conduct the statistical analysis for the annotation distributions in L3D-2K. As illustrated in Fig. 3, most of the selected keyframes encompass all categories of landmark annotations, whereas the remaining keyframes contain at least two categories. Due to the continuous variation of the laparoscopic perspective during surgery, it is unfeasible to have all types of landmarks appear in every frame. The number of labels for ligament, ridge, and silhouette are 1,817, 1,919, and 1,996, respectively, demonstrating that L3D-2K does not suffer from

class imbalance. We display some typical samples in Fig. 5. We can see that L3D-2K contains keyframes under various laparoscopic perspectives and high-quality landmark annotations.

We also investigate the relationships of all patient cases among various attributes, including gender, tumor location, tumor size, number of focuses, and cirrhosis. This information helps analyze the cohort characteristics of L3D-2K. The complete attribute descriptions are summarized in Table 1. Furthermore, we count the co-attribute distribution in Fig. 4, showing that L3D-2K spreads a diverse patient cohort with varying symptoms of liver tumors.

4. Methodology

Fig. 6 outlines the architecture of the proposed D²GPLand+. Initially, our model takes RGB keyframes as inputs and generates depth maps with an off-the-shelf depth estimation network, *i.e.*, Depth Anything V2 (Yang et al., 2024), as auxiliary inputs for depth enhancement. Then, we employ a ResNet-34 encoder (He et al., 2015) for RGB spatial feature extraction in conjunction with a frozen SAM encoder (Kirillov et al., 2023) for global geometric cue acquisition, obtaining RGB feature F_{rgb} and depth feature F_d . Subsequently, F_d is passed into our Depth-aware Prompt Embedding (DPE) module to extract geometric attributes under the guidance of landmark prompts and then generate the class-aware geometric features $F_G^{s,l,r}$. In parallel, the Cross-dimension Unified Mamba (CUMamba) block is designed to effectively integrate F_{rgb} and F_d into a unified feature representation. Further, we integrate geometric features $F_G^{s,l,r}$ with RGB-D features to produce integrated feature F_f and use a CNN decoder with residual connections to generate detection maps. Additionally, we strengthen the perception of landmark structures by adjusting the edge granularity in both implicit and explicit manners with the proposed Anatomical Feature Augmentation (AFA) scheme. Integrating all these key components into our framework ensures a comprehensive feature strengthening, including class-aware geometric learning from DPE, multi-modal fusion from CUMamba blocks, and topological emphasis from AFA scheme. As evidenced by the visualization in Fig. 6, this enables precise landmark feature extraction and significantly boosts detection performance. The following subsections will elaborate on the key components of D²GPLand+.

4.1. Depth-aware prompt embedding

To harness the advantages of pre-trained foundation models while minimizing the training costs, we maintain the SAM encoder in a frozen state. To extract geometric features associated with our task, additional fine-tuning is still required to adapt to the intraoperative domain. To this end, we randomly initialize three class-specific geometric prompts in conjunction with our DPE module to extract geometric features relevant to different landmark categories. As illustrated in Fig. 6, we initially reshape the input F_d and perform matrix multiplication with landmark prompts, generating spatial attention maps that emphasize regions associated with specific categories. Then, element-wise multiplication is performed between each attention map and the reshaped depth features, incorporating a residual operation with convolutional blocks to produce class-activated geometric features $F_G^{s,l,r}$.

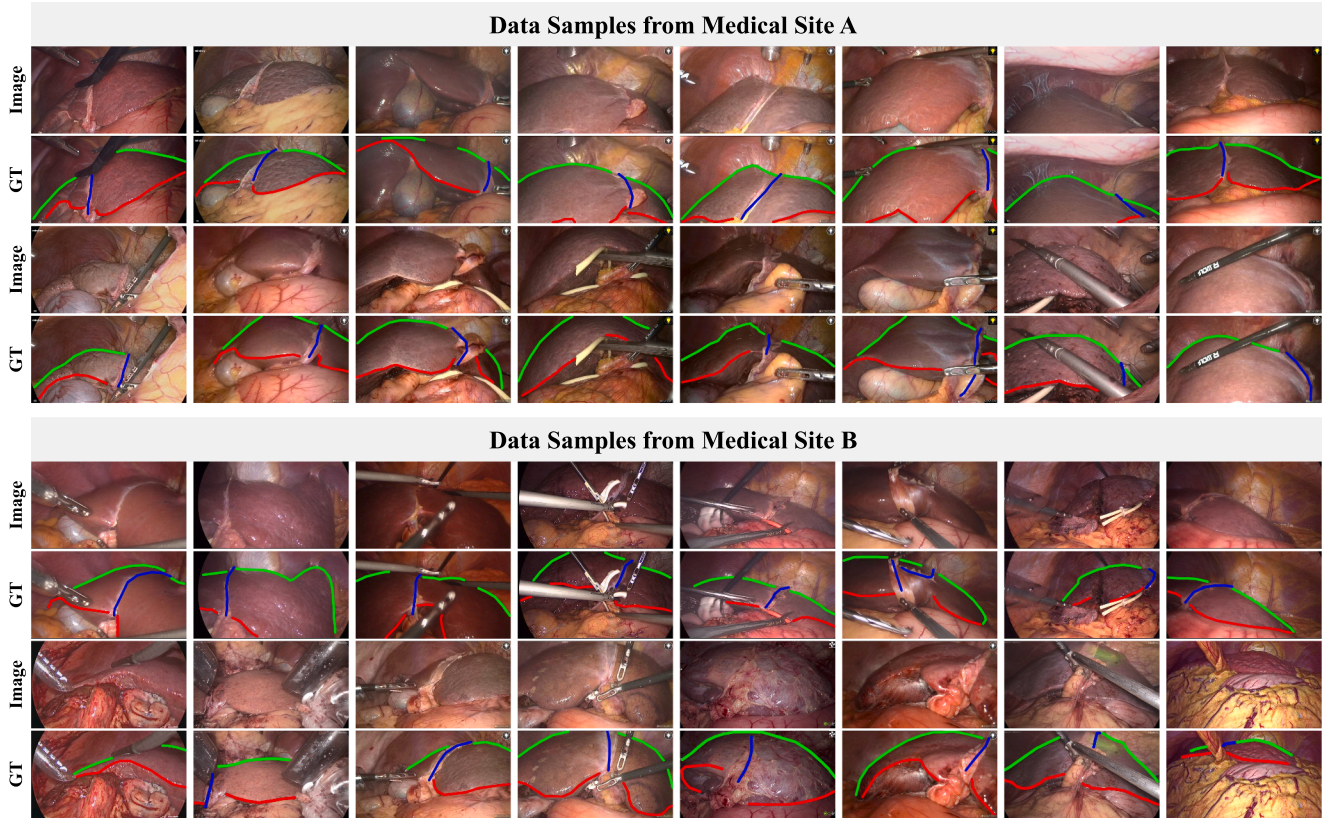


Fig. 5. Sample display of two medical sites in L3D-2K. Silhouette in green, ridge in red, and ligament in blue.

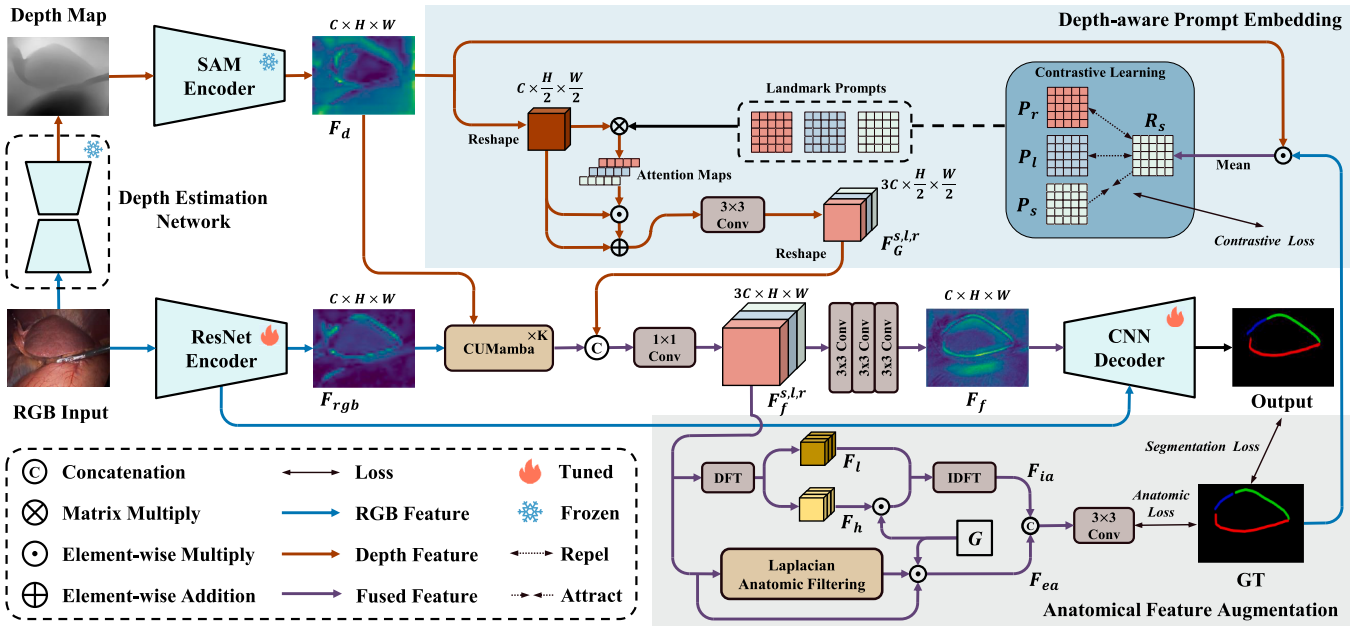


Fig. 6. Overview of the proposed D²GPLand+. Our framework contains three key components: Depth-aware Prompt Embedding (DPE) for class-aware geometric learning, Cross-dimension Unified Mamba (CUMamba) for RGB-D fusion, and anatomical Feature Augmentation (AFA) for anatomical perception. s, l, r denote three types of landmarks: silhouette, ligament, and ridge.

Due to the highly similar characteristics of liver landmarks, it still remains challenging to learn the prompts when relying solely on the indirect supervision of ground truth. To enhance the distinctiveness of prompts for effective guidance, we adopt a contrastive learning scheme as direct supervision for prompt learning. For instance, given the ground truth of the silhouette landmark and F_d , we first perform element-wise

multiplication, followed by computing the channel-wise mean to obtain the corresponding reference embeddings R_s . In the latent space, our goal is to position R_s such that it repels ridge prompt P_r and ligament prompt P_l while attracting silhouette prompt P_s . Similarly, the other two reference embeddings are supposed to attract their corresponding landmark prompts and repel the others. Upon positioning all reference

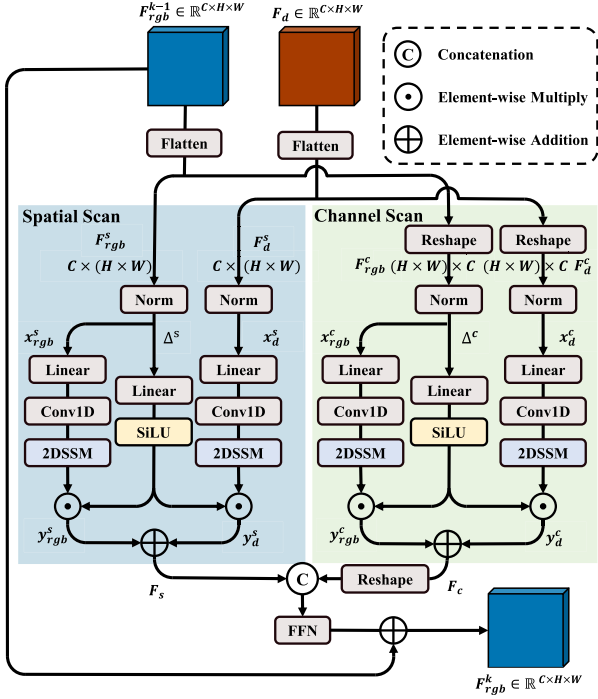


Fig. 7. Detailed depiction of the proposed CUMamba block in D²GPLand+.

embeddings, we modify the NT-Xent Loss (Chen et al., 2020) as the contrastive loss:

$$\mathcal{L}_{cl} = \frac{1}{N} \sum_{l \in L} \log \frac{\exp(P_l \cdot R_l / \tau)}{\sum_{k \in L} \exp(P_l \cdot R_k / \tau)}, \quad (1)$$

where $N = 3$ is the number of classes, $L = \{s, l, r\}$ denotes the set of landmark categories, and $\tau = 0.7$ refers to the temperature-scaled parameter. This contrastive learning strategy enhances the distinctiveness of class-specific geometry prompt representations, thereby improving the performance of extracting class-aware geometric features.

4.2. Cross-dimension unified Mamba

Inspired by cross-modal Mamba (Li et al., 2024), we propose the Cross-dimension Unified Mamba (CUMamba) module to efficiently exploit the state space model (SSM) to capture global information, as described in Fig. 7. Compared to the SE Blocks and average pooling operations in D²GPLand (Pei et al., 2024), CUMamba provides a more comprehensive RGB-D fusion based on the Mamba operation by cross-modal interplay in the spatial dimension and weighting reparameterization in the channel dimension. Suppose we embed K CUMamba blocks in the structure, the k^{th} block takes a depth feature $F_d \in \mathbb{R}^{C \times H \times W}$ and the output RGB feature of the previous layer $F_{rgb}^{k-1} \in \mathbb{R}^{C \times H \times W}$ or the original RGB feature F_{rgb} extracted by the CNN encoder as the input when $k = 1$. CUMamba consists of two separate modules: Spatial Scan and Channel Scan. The Spatial Scan is adopted to aggregate the complementary RGB as well as depth features and perform cross-modal interaction to unify holistic landmark representations. In this operation, we flatten the inputs to obtain the spatial input F_d^s and F_{rgb}^s . After that, launched from vanilla Mamba operations, we project F_{rgb}^s to modulate the gating factor Δ^s and employ Δ^s to generate the RGB spatial feature y_{rgb}^s and depth spatial feature y_d^s by element-wise multiplication, and then sum them to obtain the spatial fusion feature F_s .

The Spatial Scan focuses solely on concrete spatial information but neglects feature weighting, which is crucial for emphasizing significant features and suppressing irrelevant ones. To overcome this limitation, we design the Channel Scan to reparameterize the weights of the fea-

ture layers. In the Channel Scan, we reshape the input features F_d and F_{rgb}^{k-1} by swapping the spatial and channel dimensions, allowing for conducting the same operations to the swapped features as the Spatial Scan to produce the channel-wise fused feature F_c . After obtaining the fused features along the single dimension, we concatenate F_s and F_c and use a feed-forward network (FFN) with residual connections to derive the depth-injected RGB feature F_{rgb}^k . As depicted in Fig. 6, we concatenate the output features F_{rgb}^k and geometric features $F_G^{s,l,r}$ from the last CUMamba block and employ a 1×1 convolution to attain the class-specific fusion features $F_f^{s,l,r}$. Afterward, we utilize a series of convolutional blocks to selectively fuse the three parts of $F_f^{s,l,r}$ to get the ultimate fused feature F_f .

4.3. Anatomical feature augmentation

To capture landmark features of the liver in complex laparoscopic scenes, we present an Anatomical Feature Augmentation (AFA) scheme to improve the sensitivity to ambiguous anatomical structures in both implicit and explicit manners. Compared to Reverse Attention (RA) in D²GPLand (Pei et al., 2024) that highlights all edge regions without distinguishing landmark-related cues, AFA explicitly couples complementary implicit and explicit pathways along with learnable granularity adjustment to enhance the perception of landmark evidence and maintain anatomical geometry. As shown in the right bottom of Fig. 6, AFA consists of two branches: the top branch represents implicit augmentation with DFT, and the bottom is the explicit augmentation through our Laplacian anatomical filtering. We then utilize learnable granularity tokens to constrain the detail levels of implicit and explicit features. Finally, we concatenate them and use convolutional blocks to generate customized anatomical features.

4.3.1. Learnable granularity adjustment

Since the characteristics of the target anatomical structures are highly similar to the surrounding tissues, we propose a learnable granularity token G , which is applied to implicit high-frequency features to control the detail levels of the anatomical representations learned by the model, thus eliminating outliers that may mislead the detection. As different types of landmarks contain varying characteristics, we take $F_f^{s,l,r}$ as input and use $G \in \mathbb{R}^{3 \times 1}$ to constrain the detail level of each anatomical structure.

Considering that high-frequency features contribute to the depiction of sudden changes, i.e., tissue edges where landmarks are located, applying the weighting strategy can better control the granularity of anatomical features. Thus, we transform class-specific fusion features $F_f^{s,l,r}$ into the frequency domain and further decompose them into low-frequency features F_l and high-frequency features F_h with DFT:

$$F_l = B \odot \text{DFT}(F_f^{s,l,r}); F_h = (\mathbb{1} - B) \odot \text{DFT}(F_f^{s,l,r}), \quad (2)$$

where $\odot$ is element-wise multiplication. $B \in \mathbb{R}^{H \times W}$ denotes the matrix with ones in the center area and zeros elsewhere. $\mathbb{1} \in \mathbb{R}^{H \times W}$ is the matrix of ones. The size of the center area of B is calculated with a frequency separation ratio c , which is set as 0.3 in our model. Then, we multiply the high-frequency features F_h by G to adjust detail richness and get F_h' . The implicitly augmented feature F_{ia} is generated by:

$$F_{ia} = \text{IDFT}(F_l + F_h' \cdot G), \quad (3)$$

where we add F_l and F_h' together and use the inverse DFT (IDFT) to transform the features back to the spatial domain.

4.3.2. Explicit Laplacian anatomical filtering

Owing to the uncontrollable learning process and the potential representation bias in implicit feature extraction, we embed another explicit Laplacian anatomical filtering module to complementarily enhance the perception of anatomical-related features by applying Gabor Wavelet

Table 2

Comparison with cutting-edge models for liver landmark detection on L3D (Pei et al., 2024), L3D-2K, and P2ILF (Ali et al., 2025) test sets. **Bold texts** denote the best result in each column, and underlined texts denote the second-best.

Methods	Pub. & Year	L3D (Pei et al., 2024)			L3D-2K			P2ILF (Ali et al., 2025)			Infer. Speed ↓	GFLOPs ↓
		DSC ↑	IoU ↑	Assd ↓	DSC ↑	IoU ↑	Assd ↓	DSC ↑	IoU ↑	Assd ↓		
Non-SAM-based methods												
UNet (Ronneberger et al., 2015)	MICCAI'15	51.39	36.35	84.94	35.17	23.74	76.78	29.89	17.89	47.95	172.65ms	774.30
COSNet (Labrunie et al., 2023)	MIDL'23	56.24	40.98	69.22	37.35	25.96	59.24	26.78	16.05	47.33	167.25ms	319.24
ResUNet (Xiao et al., 2018)	ITME'18	55.47	40.68	70.66	40.39	28.43	55.80	25.84	15.76	50.97	155.76ms	314.65
UNet++ (Zhou et al., 2019)	TMI'19	57.09	41.92	74.31	38.70	27.26	58.70	34.96	21.57	42.39	213.19ms	554.87
HRNet (Wang et al., 2020)	TPAMI'20	58.36	43.50	70.02	41.19	29.36	56.40	33.64	21.31	44.16	146.82ms	376.52
DeepLabv3+ (Chen et al., 2018)	ECCV'18	59.74	44.92	60.86	41.55	29.89	55.11	27.62	16.42	50.77	51.05ms	105.45
TransUNet (Chen et al., 2021)	arXiv'21	56.81	41.44	76.16	37.54	25.85	46.75	25.49	15.22	52.14	262.21ms	795.10
SwinUNet (Cao et al., 2022)	ECCV'22	57.35	42.09	72.80	36.32	24.89	48.23	18.65	10.62	68.46	<u>127.22ms</u>	<u>129.68</u>
CaseNet (Yu et al., 2017)	CVPR'17	60.94	46.09	63.89	40.23	28.37	56.55	33.94	21.33	45.02	233.68ms	819.27
Unet 3+ (Huang et al., 2020)	ICASSP'20	56.61	40.96	62.03	38.59	27.18	60.12	30.57	18.69	46.65	194.09ms	253.27
double ResUNet (Ali et al., 2025)	MedIA'25	58.39	43.52	63.78	40.53	28.60	53.73	36.54	22.80	39.26	342.47ms	682.53
nnU-Net (Isensee et al., 2024)	MICCAI'24	60.97	46.14	62.03	41.25	29.30	51.28	35.41	22.06	42.64	237.06ms	791.52
SAM-based methods												
SAM-Adapter (Chen et al., 2023)	ICCV'23	57.57	42.88	74.31	39.20	26.81	59.61	21.12	12.00	57.13	300.21ms	489.35
SAMed (Zhang and Liu, 2023)	arXiv'23	62.03	47.17	61.55	43.72	31.20	43.01	31.73	19.42	40.08	278.56ms	488.95
SAM-LST (Chai et al., 2024)	PCS'24	60.51	45.03	68.87	39.43	27.39	56.69	28.75	18.38	46.53	304.78ms	517.68
AutoSAM (Hu et al., 2023)	arXiv'23	59.12	44.21	62.49	40.47	28.57	54.58	25.71	15.43	53.65	337.44ms	589.21
D ² GPLand (Pei et al., 2024)	MICCAI'24	63.52	48.68	59.38	49.31	35.76	41.34	40.55	25.87	38.73	297.93ms	572.85
D ² GPLand + (Ours)	–	64.62	50.14	57.68	51.66	37.59	36.24	42.15	27.40	37.12	288.84ms	551.19

filtering (Lee, 1996) with the Laplacian scaling function. The Gabor Wavelet in 2D space is defined as

$$g(x, y; \lambda, \theta, \psi, \sigma, \gamma) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cdot \exp\left(i\left(\frac{2\pi}{\lambda}x' + \psi\right)\right), \quad (4)$$

where $x' = x\cos\theta + y\sin\theta$, $y' = -x\sin\theta + y\cos\theta$, λ is the wavelength, θ is the orientation angle of the Gabor kernel, ψ is the phase offset, σ is the standard deviation of the Gaussian function, and γ is the aspect ratio between the spatial width and height. To highlight the anatomical features, we then employ the Laplacian operator Δf as the scaling function, which improves the discrimination of the boundary areas. The Laplacian operator in two-dimension space is described as

$$\Delta f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}. \quad (5)$$

Therefore, the Gabor Wavelet filtering with Laplacian scaling function can be formulated as

$$g_L(x, y) = \Delta f(x, y) \cdot g(x, y), \quad (6)$$

where we remove the parameters for brevity. To capture information from various orientations, we equally sample 18 θ values to generate filter kernels and obtain the specific features with convolutions. We employ G to adjust the edge granularity to obtain the explicitly augmented features F_{ea} :

$$F_{ea} = G \cdot \left(\sum_{i=1}^A g_L(x, y; \theta_i) \circ F_f^{s, i, r}\right), \quad (7)$$

where $A = 18$ denotes the total number of sampled θ values and $\circ$ is the convolution operation. Finally, we concatenate F_{ia} and F_{ea} and employ convolutional blocks to extract the anatomical features. Here, we combine Dice loss and BCE loss as the anatomical Loss $\mathcal{L}_{ana}$ to supervise model learning.

4.4. Loss function

In addition to the above-mentioned contrastive loss and anatomical loss, we also add the segmentation loss $\mathcal{L}_{seg}$ to the overall loss function to supervise the final detection map:

$$\mathcal{L}_{seg} = \frac{1}{N} \sum_{l \in L} (\mathcal{L}_{dice}^{(l)} + \mathcal{L}_{bce}^{(l)}), \quad (8)$$

where $\mathcal{L}_{dice}^{(l)}$ denotes the Dice Loss and $\mathcal{L}_{bce}^{(l)}$ denotes the binary cross-entropy (BCE) loss. In summary, the total loss function can be defined as

$$\mathcal{L}_{total} = \lambda_{seg} \mathcal{L}_{seg} + \lambda_{cl} \mathcal{L}_{cl} + \lambda_{ana} \mathcal{L}_{ana}, \quad (9)$$

where λ_{seg} , λ_{cl} , and λ_{ana} are the balancing parameters for $\mathcal{L}_{seg}$, $\mathcal{L}_{cl}$, and $\mathcal{L}_{ana}$, respectively. They are empirically set to 1 for optimal performance.

5. Experiments

5.1. Experimental setup

5.1.1. Datasets

In this study, we conduct experiments on three laparoscopic liver landmark detection datasets for evaluations, including two publicly available datasets L3D (Pei et al., 2024) and P2ILF (Ali et al., 2025), and the proposed L3D-2K dataset. All datasets contain three liver landmark categories to be detected: ridge, ligament, and silhouette. L3D consists of 1152 annotated keyframes of 1920 × 1080 pixels from laparoscopic liver surgical videos, where 921 keyframes are used for training, 122 for validation, and 109 for testing. P2ILF has 183 annotated images with the same landmark categories as L3D for liver landmark detection. Wherein, 167 images are used for training while the others are used for testing. Since only the training set is available, we randomly select 124 images for training and the remaining 43 images for testing in our experiments. The detailed information of L3D-2K can be found in Section 3.1.

5.1.2. Evaluation metrics

Referring to other works on medical image segmentation tasks (Pei et al., 2024; Zhang et al., 2024; Wang et al., 2025), we select the commonly used Jaccard Index (IoU) and Dice Score Coefficient (DSC) for comprehensive evaluation. DSC tends to evaluate the average performance of models, while IoU tends to evaluate the worst-case performance. Both of them are reported in percentages (%). Meanwhile, we utilize Average Symmetric Surface Distance (Assd), reported in pixels, to measure the minimum distance between the predicted and actual values for the boundary alignment problem. We also report inference speed in milliseconds (ms) per keyframe and computational cost in giga floating-point operations (GFLOPs) at the input resolution.

Table 3
Comparison with P2ILF challenge methods on L3D, L3D-2K, and P2ILF test sets.

Methods	L3D			L3D-2K			P2ILF			Infer. Speed↓	GFLOPs↓
	DSC↑	IoU↑	Assd↓	DSC↑	IoU↑	Assd↓	DSC↑	IoU↑	Assd↓		
BHL (Ali et al., 2025)	58.39	43.52	63.78	40.53	28.60	53.73	36.57	22.80	39.26	342.47ms	682.53
UCL (Ali et al., 2025)	57.09	41.92	74.31	38.70	27.26	58.70	34.96	21.57	42.39	<u>213.19ms</u>	554.87
VIP (Ali et al., 2025)	53.71	38.32	70.59	34.26	22.87	68.09	26.14	15.72	54.45	204.17ms	389.73
VOR (Ali et al., 2025)	50.62	35.79	77.51	33.16	21.64	78.60	24.65	14.58	59.20	571.26ms	1106.84
NCT (Ali et al., 2025)	<u>60.97</u>	<u>46.14</u>	<u>62.03</u>	<u>41.25</u>	<u>29.30</u>	<u>51.28</u>	35.41	22.06	40.64	237.06ms	791.52
D²GPLand+	64.62	50.14	57.68	51.66	37.59	36.24	42.15	27.40	37.12	288.84ms	<u>551.19</u>

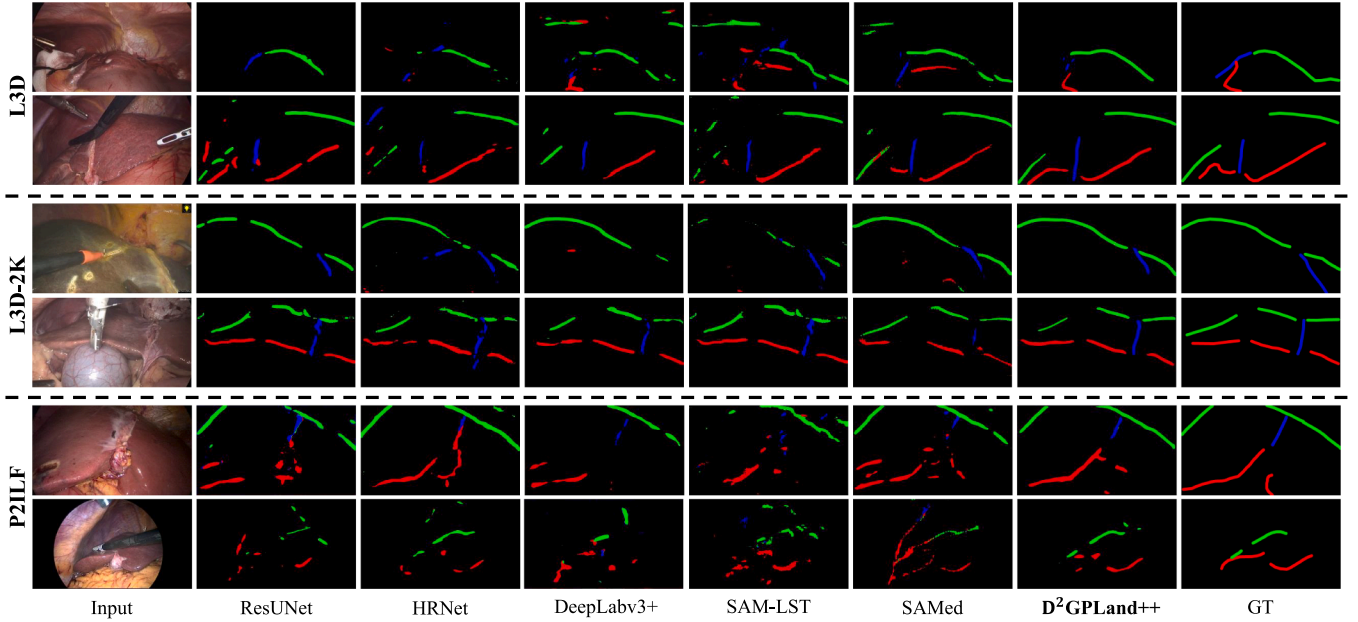


Fig. 8. Visualizations of D²GPLand+ and other well-performed methods on L3D (Pei et al., 2024), L3D-2K, and P2ILF (Ali et al., 2025) test sets.

5.1.3. Implementation details

The training and testing processes of D²GPLand+ are executed on a single NVIDIA RTX A6000 GPU. We run 60 epochs for training with a batch size of 2. A frozen pre-trained SAM-b (Kirillov et al., 2023) is implemented in the depth encoder. We set $K = 5$ to embed five CUMamba blocks in our architecture. We first resize all images to 1022×1022 tailored to the frozen Depth Anything Model and get the corresponding depth maps with the same size. Then, the images and depth maps are resized to 1024×1024 for the SAM encoder. We further apply random rotation and flip for data augmentation. The Adam optimizer is used with the initial learning rate of $1e-4$ and weight decay factor of $3e-5$. Additionally, the CosineAnnealingLR scheduler is applied to adjust the learning rate to $1e-5$.

5.2. Experimental results

5.2.1. Comparison with state-of-the-art methods

We compare our D²GPLand+ with 17 cutting-edge methods on L3D, L3D-2K, and P2ILF test sets. These methods are divided into two types: 1) Non-SAM-based models, including UNet (Ronneberger et al., 2015), COSNet (Labrunie et al., 2023), ResUNet (Xiao et al., 2018), DeepLabV3+ (Chen et al., 2018), UNet++ (Zhou et al., 2019), HRNet (Wang et al., 2020), TransUNet (Chen et al., 2021), SwinUNet (Cao et al., 2022), CaseNet (Yu et al., 2017), Unet 3+ (Huang et al., 2020), double ResUNet (Ali et al., 2025), and nnU-Net (Isensee et al., 2024), and 2) SAM-based models, including SAM-Adapter (Chen et al., 2023), SAMed (Zhang and Liu, 2023), AutoSAM (Hu et al., 2023), SAM-LST (Chai et al., 2024), and D²GPLand (Pei et al., 2024). For a fair comparison, all competitors are trained to converge with official imple-

mentations. The quantitative comparisons are presented in Table 2. It is observed that D²GPLand+ outperforms competitors across all metrics on three datasets and statistical analysis confirms that these improvements are significant (p -value < 0.05). On the L3D dataset, compared to the top-ranked SAMed, our method improves 2.59% in DSC, 2.97% in IoU, and 3.87 pixels in Assd metrics, demonstrating the effectiveness of incorporating depth cues into the RGB modal for landmark detection. Besides, we observe that non-SAM-based methods exhibit inferior performance compared to most SAM-based methods. It illustrates that the global geometric feature from the pre-trained SAM encoder can enhance landmark perception. Notably, on the L3D-2K dataset, all methods experience a decline in performance compared to the L3D dataset. It may be attributed to the inclusion of more practical but challenging cases to L3D-2K (see Section 3.1.1). Nevertheless, our approach still outperforms other methods by a large margin. Similarly, as shown in the results on the P2ILF dataset, our method also achieves the best detection results with a DSC of 42.15%, IoU of 27.40%, and Assd of 37.12 pixels. Compared to SAM-based models, D²GPLand+ is second only to SAMed in inference speed and superior to the previous version (Pei et al., 2024). Fig. 8 also exhibits the visualization results on all three datasets. We can see that our method provides more accurate detection while mitigating the impact of occlusion from other tissues and instruments.

5.2.2. Comparison with P2ILF challenge methods

In addition, we also compare with the methods that have achieved top performance in the P2ILF challenge. According to Ali et al. (2025), we select the methods from BHL, NCT, UCL, VIP, and VOR teams for comparison. Since the official code is unavailable, we reimplement the methods based on the descriptions in Ali et al. (2025). As shown in Ta-

Table 4
Ablations for key components of our D²GPLand+.

Models	CUMamba	DPE	AFA	DSC ↑	IoU ↑	Assd ↓
M.1				37.69	27.13	58.76
M.2	✓			40.25	28.68	55.38
M.3	✓		✓	41.56	30.15	52.41
M.4		✓	✓	41.87	30.68	52.25
M.5	✓	✓	✓	42.13	31.22	48.57
Ours	✓	✓	✓	43.65	32.08	45.19

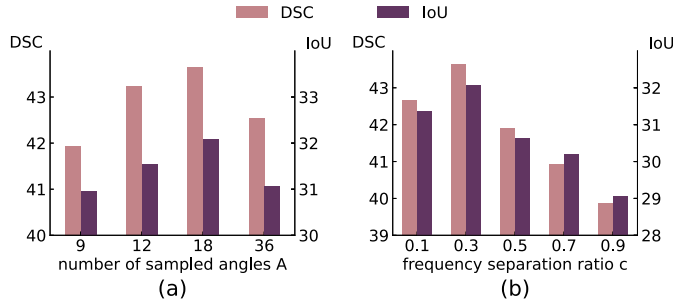


Fig. 9. Ablations for AFA module. (a): the number of sampled angles A . (b): different frequency separation ratio c .

ble 3, our model delivers superior detection accuracy across all three datasets with the performance gains being statistically significant (p -value < 0.05). On L3D and L3D-2K test sets, D²GPLand+ consistently surpasses the second-best NCT method by a large margin, especially lowering Assd by 15.04 pixels on L3D-2K, which suggests notably improved geometric consistency and boundary adherence of our method under complex conditions. On P2ILF test set, D²GPLand+ also significantly outperforms the P2ILF challenge methods. In terms of efficiency, our method maintains the second-lowest computational cost (551.19 GFLOPs) and a competitive inference time for one frame (288.84ms).

5.3. Detail ablation analysis for D²GPLand+

We conduct a series of ablation experiments on the L3D-2K validation set to evaluate and analyze the effectiveness of our framework. We perform significance tests for all ablation experiments. The results demonstrate that the improvements attributed to our optimized framework are statistically significant (p -value < 0.05).

5.3.1. Analysis of key components

Table 4 shows the contribution of each key component in D²GPLand+. The baseline (M.1) comprises a ResNet-34 encoder and frozen SAM-b encoder, and we directly concatenate RGB and depth features before feeding them into the CNN decoder. Overall, each component contributes to the performance of our framework. M.2 and M.4 indicate the effectiveness of the CUMamba block in merging RGB and depth features. Based on the M.2 variant, M.5 combined with our DPE module to further enhance the performance by incorporating geometry clues into RGB features. M.3 and M.2 demonstrate the effectiveness of enhancing the perception of the anatomical structures after adding the AFA scheme. By assembling all components, we obtain a significant improvement over the baseline, i.e., 5.96% in DSC, 4.95% in IoU, and 13.57 pixels in Assd metric.

5.3.2. Ablations for AFA module

We study the effect of the number of sampled angles A and the frequency separation ratio c in our AFA module. More orientation angles provide more high-frequency features but also embrace features that are not associated with anatomical features. As shown in Fig. 9(a), we explore the sampled angles by changing $A = \{9, 12, 18, 36\}$, which separates one cycle with 20, 15, 10, and 5 degrees, respectively. It can be

Table 5

Ablations for key components of AFA module. LGA denotes learnable granularity adjustment, IA denotes implicit augmentation, and ELAF denotes explicit Laplacian anatomical filtering.

Models	LGA	IA	ELAF	DSC ↑	IoU ↑	Assd ↓
A.1				42.13	31.22	48.57
A.2	✓	✓		43.19	31.72	47.70
A.3	✓		✓	42.58	31.53	45.84
A.4			✓	42.36	31.35	45.89
Ours	✓	✓	✓	43.65	32.08	45.19

Table 6

Comparison of anatomical feature enhancement modules.

Methods	DSC ↑	IoU ↑	Assd ↓
Baseline	42.13	31.22	48.57
Reverse Attention (Pei et al., 2024)	42.56	31.47	46.32
AFA (Ours)	43.65	32.08	45.19

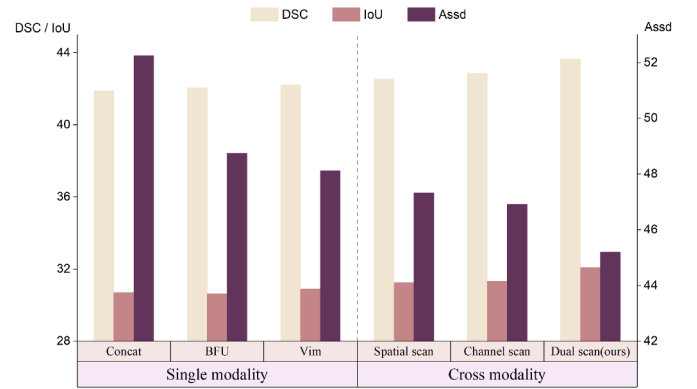


Fig. 10. Ablations for the CUMamba block in D²GPLand+.

observed that the performance improves when A increases from 9 to 18 and decreases when A is equal to 36. Hence, we set A to 18 for the optimal performance. In parallel, the value of c determines the boundary of high- and low-frequency features, where a higher c value allows controlling more details of features while a lower one affects fewer features. We also verify the effect of AFA when the value of c ranges from 0.1 to 0.9 in steps of 0.2. Fig. 9(b) illustrates that lower c values ($c < 0.5$) achieve better results by controlling the detail levels of features with more sudden frequency changes. However, when the value of c is too small (e.g., $c = 0.1$), anatomical characteristics may not be fully captured. According to the ablation results, we set c to 0.3 by default to separate high- and low-frequency features.

We additionally analyze the individual contribution of the key components in the AFA module, including the Learnable Granularity (LGA), Implicit Augmentation (IA), and Explicit Laplacian Anatomical Filtering (ELAF). As described in Section 4.3, LGA is used to modulate the detail level of F_{ia} and F_{eq} , so we combine LGA with IA or with ELAF rather than evaluating LGA separately. Likewise, IA is grouped with LGA to properly shape the high-frequency features. As reported in Table 5, the complete AFA module achieves the best performance on all metrics, and each key component contributes to the detection performance. Concretely, we can obtain that ELAF brings more improvements in Assd than IA by comparing A.2, A.3, highlighting its powerful capability of topology preservation. In contrast, IA contributes more to enhancing the capture of edge features, resulting in higher DSC and IoU performance. Besides, A.3 and A.4 show that LGA assists ELAF in mitigating representation bias, although its primary advantage lies in suppressing outliers in high-frequency features. By jointly integrating all components, the AFA module produces a remarkable improvement over the baseline.

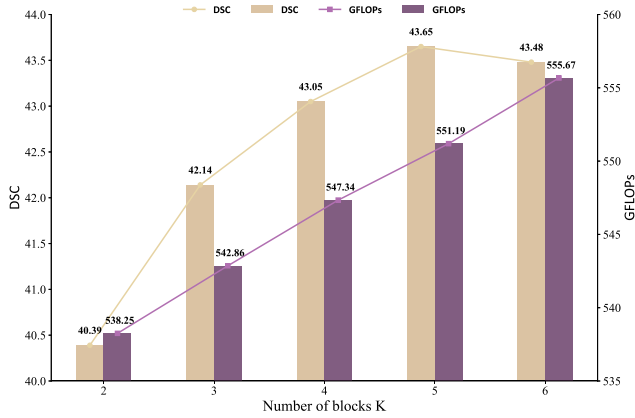


Fig. 11. Ablations for the number of CUMamba blocks K .

We further compare our AFA module with the Reverse Attention (RA) mechanism in previous work (Pei et al., 2024) to assess its benefits on detection performance. As illustrated in Table 6, AFA achieves the highest detection accuracy, surpassing both baseline and RA. Compared to baseline, whose setting is identical to M.5 in Table 4, AFA improves DSC by 1.52% and IoU by 0.86%, which is approximately three times larger than the improvements achieved by RA (0.43% and 0.25% in DSC and IoU, respectively). Meanwhile, AFA also delivers a large improvement of 1.13 pixels in Assd than RA. These results indicate that AFA enhances model performance more substantially than the RA mechanism.

5.3.3. Ablations for CUMamba

We further conduct ablations to evaluate the CUMamba block for RGB-D feature fusion across spatial and channel dimensions. We compare five variants with CUMamba in Fig. 10: 1) Concat, simple concatenation of RGB-D features, 2) BFU, the bi-modal feature unification module from D²GPLand (Pei et al., 2024), 3) Vim, Vision Mamba (Zhu et al., 2024) applied separately to each modality before concatenation, 4) Spatial scan, retaining only spatial scan, and 5) Channel scan, applying only channel scan. These strategies are grouped into two categories: ‘Single modality’ and ‘Cross modality’, denoting whether cross-modal interaction occurs during fusion. The results show that the methods in ‘Single modality’ perform worse than the ones in ‘Cross modality’, highlighting the importance of cross-modal interactions. The better performance of our Dual Scan demonstrates the necessity of combining both spatial- and channel-wise feature fusion. In addition, we evaluate the efficiency of CUMamba in Table 9. Both detection performance and efficiency of the BFU module is inferior to that of the proposed CUMamba. Compared to the self-attention operation in Transformer (Vaswani, 2017), Mamba has a significant advantage in terms of computational complexity. Overall, CUMamba achieves a higher detection accuracy with faster inference speed and lower computational complexity. We also conduct experiments to evaluate the influence of various number of stacked CUMamba blocks (K). As illustrated in Fig. 11, there is an obvious trend that the detection performance improves when K increases from 2 to 5 and peaks at $K = 5$. However, further increasing from 5 to 6 leads to a slightly worse performance, while the inference speed and computational complexity keep rising. To this end, to achieve the balance between accuracy and efficiency, we default to the choice of $K = 5$ in our architecture.

5.3.4. Backbone selections

To explore the effect of various combinations of backbones across RGB and depth modalities, we conduct ablations with different encoder combinations (i.e., ResNet-34 He et al., 2015 and SAM-b Kirillov et al., 2023) in our framework. As shown in Table 7, D²GPLand + achieves superior performance when leveraging the ResNet-34

Table 7

Ablations for backbone combinations on different modalities. R and D denote the RGB and depth modalities.

Methods	Backbones	DSC ↑	IoU ↑	Assd ↓
Dual CNN	ResNet-34 (R & D)	41.73	30.43	51.34
Dual SAM	SAM (R & D)	42.34	30.89	49.23
SAM + CNN	ResNet-34(D) + SAM(R)	42.92	31.24	46.62
SAM + CNN	R-34(R) + SAM(D)	43.65	32.08	45.19

Table 8

Effect of geometric prompt learning.

Methods	DSC ↑	IoU ↑	Precision ↑	Recall ↑
w/o prompt learning	41.56	30.35	41.20	38.79
prompt learning (Ours)	43.65	32.08	42.58	44.97

Table 9

Comparison of RGB-D fusion strategies.

Methods	DSC ↑	Infer. speed ↓	GFLOPs ↓
BFU (Pei et al., 2024)	42.06	302.56ms	568.63
Self-Attention (Vaswani, 2017)	42.71	394.03ms	748.55
CUMamba (Ours)	43.65	288.84ms	551.19

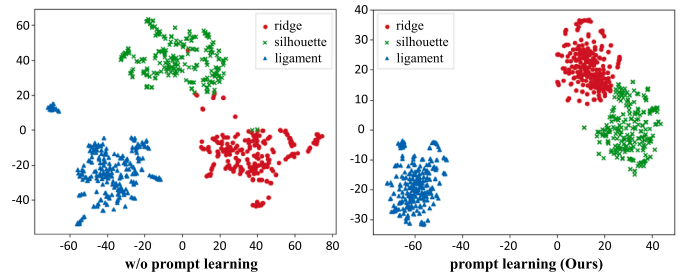


Fig. 12. t-SNE of features w/ and w/o depth prompt learning.

encoder for RGB frames and the SAM encoder for depth modality. It indicates that the ResNet-34 encoder is sensitive in capturing lower-level anatomical structures, while SAM excels in extracting global geometric features. The RGB frames contain detailed lower-level features for anatomical structure identification but also include noise factors, while depth maps provide global shape characteristics and clean geometric insights. The ablation result demonstrates that the SAM encoder in our framework exhibits higher sensitivity to global geometric features in the depth modality.

5.3.5. Effects of prompt learning

We evaluate the effect of our geometric prompt learning in DPE. As shown in Table 8, with prompt learning, the model performance gains 2.09% in DSC and 1.73% in IoU. We also employ Precision and Recall metrics that can better indicate the accuracy of positive samples across different landmark categories. The results indicate that by applying prompt learning, the model achieves a slightly better performance in Precision and a significantly higher performance in Recall, demonstrating that our prompt learning strategy imbues the network with a better ability to distinguish different landmarks. In addition, we visualize the pixel representations learned from two variants with t-SNE in Fig. 12. The features from different categories converge into more compact clusters with prompt learning. Notably, silhouette and ridge clusters are located closer than the ligament cluster in the t-SNE results of our model, while ‘w/o prompt learning’ yields uniformly spaced clusters. Since silhouettes and ridges have more similarities than ligaments, this observation further proves the validity of our strategy.

Table 10
Ablations for incorporating the depth modality.

Modalities	Methods	DSC $\uparrow$	IoU $\uparrow$	Assd $\downarrow$
w/o Depth	–	37.51	27.37	62.78
Depth	Adelai Depth (Yin et al., 2022)	41.61	30.34	51.39
Depth	DAMv2 (Yang et al., 2024)	43.65	32.08	45.19

5.3.6. Effects of depth modality

To investigate the impact of the depth modality on our model, we compared the default DAMv2 (Yang et al., 2024) with two variants: removing the estimated depth map from the input and replacing it with the Adelai Depth (Yin et al., 2022) used in D²GPLand (Pei et al., 2024). As illustrated in Table 10, removing the depth input leads to considerable performance degradation, confirming the crucial role of depth features. Compared to the depth map inferred by Adelai Depth, the default setting obtains more accurate depth information, thereby improving the accuracy of landmark detection.

5.3.7. Ablations for loss function and learning rate

We conduct comprehensive experiments to study the effects of different choices of training hyperparameters following the experimental settings in a recent work (Borgars et al., 2025). This ablation experiment includes three groups of experiments: 1) Loss function ablations, in which we evaluate the effectiveness of each loss component, L_{seg} , L_{cl} , and L_{ana} , by removing them from our final loss function, 2) Loss weight ablations, where we compare the performance of three pairs of loss weights of $(\lambda_{seg}, \lambda_{cl}, \lambda_{ana} \in (2, 0.5, 0.5), (0.5, 2, 0.5), (0.5, 0.5, 2))$ and our balanced loss weights (1, 1, 1), and 3) Learning rate ablations that we apply four different initial learning rates of $1e-2$, $1e-3$, $1e-4$, $5e-5$ to train the model. The results are summarized in Table 11. The top part reports the results of the ablations for loss components. We can obtain that removing any loss component degrades the detection performance relative to using full loss function, except at the learning rate of $1e-2$, owing to exploding gradients (see below). The bottom part of Table 11 represents the loss weights ablation. The balanced loss weights (1,1,1) consistently yield the best detection performance compared to the other candidates. Last but not least, varying the learning rate shows a clear optimum at $1e-4$. Specifically, the performance of all methods improves when reducing the learning rate from $1e-2$ to $1e-4$ and drops when further decreasing to $5e-5$. Notably, with an initial learning rate of $1e-2$, we find that all candidates that using full loss (the bottom part in Table 11) suffer from exploding gradients, where the loss values become infinite after several epochs during training, leading to markedly inferior performance compared to other variants in Loss function ablations. This finding demonstrates that applying a smaller initial learning rate is preferable when optimizing our full loss function. Overall, the results indicate that integrating all loss components with balanced weights (1,1,1) and a learning rate of $1e-4$ provides the most favorable performance and model stability.

5.3.8. Ablations for model generalizability

To evaluate the robustness and generalization capabilities of our proposed D²GPLand+, we conduct a comprehensive analysis using L3D-2K and P2ILF datasets. We compare our model against four leading methods, including SAMed (Zhang and Liu, 2023), SAM-LST (Chai et al., 2024), CaseNet (Yu et al., 2017), and DeepLabV3 (Chen et al., 2018). As presented in Table 12, with the aim of assessing the models' adaptability to an unseen, smaller dataset, we design four distinct experimental settings for comprehensive evaluation: 1) Baseline, where models are trained and tested exclusively on the P2ILF dataset, 2) Zero-shot, where models are trained on L3D-2K training set and then directly tested on P2ILF test set to evaluate their generalization capability, 3) 50% Fine-tuning, where models pre-trained on L3D-2K training set are subsequently fine-tuned with half of the P2ILF training data and then

tested on P2ILF test set, simulating the clinical scenarios with very limited domain-related data, and 4) Fully Fine-tuning, where models pre-trained on L3D-2K are fine-tuned with the entire P2ILF training set.

Table 12 illustrates that D²GPLand+ consistently outperforms other approaches by a large margin across all four experimental settings. Specifically, the baseline experiment demonstrates the efficacy of our design in low-data scenes. The zero-shot setting highlights the powerful capability of our model in learning generalizable priors. After pre-training on the larger L3D-2K dataset, our model's zero-shot performance not only surpasses all competing methods but also significantly exceeds its own baseline performance, confirming that the prior knowledge learned by our model is generalizable. The results of 50% fine-tuning and full fine-tuning further underscore this conclusion. With only 50% of the P2ILF training data for fine-tuning, our model obtains a strong performance gain of 11.81% in DSC over the baseline. When fully fine-tuned, D²GPLand+ reaches a state-of-the-art performance, establishing a significant margin over the second-rank model, DeepLabv3+, of 5.67%, 4.24%, and 2.82 pixels in DSC, IoU, and Assd, respectively. In summary, this ablation analysis strongly corroborates the superior generalization capabilities of our framework. Further, following L3D-2K training, significant performance improvements are obtained across all models in zero-shot and fine-tuning settings, demonstrating the substantial contribution of our dataset and high potential value for advancing related research and clinical practice.

6. Discussion

Landmarks are essential biomarkers for enabling 2D-3D fusion and facilitating AR navigation, as they provide surgeons with consistent location information (Adagolodjo et al., 2017; Koo et al., 2017, 2022). In typical laparoscopic liver surgery, surgeons localize liver landmarks and perform surgical operations largely based on their personal intraoperative experience. Such uncertainty in landmark localization can lead to suboptimal or erroneous actions like unexpected parenchymal resection. Employing an automatic landmark detection approach provides surgeons with precise intraoperative visual guidance to support appropriate surgical decisions, thereby potentially reducing surgical risk and shortening the hospital stay of patients. In addition, liver landmarks can serve as reference markers for aligning consistent preoperative 3D anatomy and complex intraoperative 2D keyframes, facilitating robust liver registration and paving the way towards AR navigation. However, due to the complex environment of laparoscopy, especially tissue deformation, existing liver landmark detection methods fall short of promising performance.

To bridge this gap, we build a dedicated dataset and present a novel approach for better liver anatomical representations. For data contribution, the proposed L3D-2K is currently the largest publicly available dataset for laparoscopic liver landmark detection.

Notably, despite L3D (Pei et al., 2024) and L3D-2K datasets collected from same medical sites, all methods show a remarkable performance gap on both datasets. This discrepancy is likely attributed to L3D-2K containing more challenging cases (see Section 3.1.1). Fig. 13 (left) shows two representative hard samples. In the top sample, the biased view of the laparoscopic camera resulted in difficulty in distinguishing the silhouette (green area) from the ridge (red area). In contrast, the difficulty in the bottom sample stems from the increased camera-to-liver distance, resulting in a smaller target area. Beyond these challenges, other anomalies, e.g., blood, smoke, and occlusion by instruments, further complicate liver landmark detection. Samples involving the above factors are supplemented in L3D-2K, increasing the difficulty of detecting and better reflecting real-world clinical scenarios.

For methodological contributions, we develop a depth-induced prompt learning framework, D²GPLand+, which embeds depth-based prompts for specific landmark categories and adopts contrastive learning to distinguish liver landmarks. Moreover, we propose the CUMamba block, which performs dual-modal scan across both spatial and channel

Table 11

Ablations for loss function and learning rate. We use **bold texts** to highlight the best results in each column and highlighted cells to represent the best results in each row. LR denotes learning rate and Weights denote values of $(\lambda_{seg}, \lambda_{cl}, \lambda_{ana})$.

Weights \ LR	1e-2		1e-3			1e-4 (Ours)			5e-5			
	DSC ↑	IoU ↑	Assd ↓	DSC ↑	IoU ↑	Assd ↓	DSC ↑	IoU ↑	Assd ↓	DSC ↑	IoU ↑	Assd ↓
Loss function ablations												
(1, 0, 0)	36.54	26.48	58.57	38.89	28.13	57.10	40.67	29.87	55.64	40.30	29.21	62.23
(1, 0, 1)	33.03	23.99	67.54	39.17	28.34	55.42	41.95	30.84	49.73	41.40	30.39	53.25
(1, 1, 0)	36.11	25.43	59.03	39.33	28.77	54.18	41.45	30.49	51.61	39.64	28.97	53.72
Loss weights ablations												
(2, 0.5, 0.5)	26.12	16.86	82.86	39.00	28.70	54.85	41.83	30.99	49.28	40.95	30.09	51.41
(0.5, 2, 0.5)	22.43	13.83	98.93	38.76	28.02	56.24	42.21	31.03	49.19	40.73	29.61	52.86
(0.5, 0.5, 2)	26.48	17.15	84.05	39.37	28.86	52.21	42.70	31.28	47.21	41.36	30.03	50.09
(1, 1, 1) (Ours)	26.81	17.87	81.42	40.09	29.49	52.13	43.65	32.08	45.19	41.66	30.44	48.70

Table 12

Ablations for model generalizability on P2ILF test set.

Methods	Settings		Baseline			Zero-shot			50% Fine-tuning			Fully Fine-tuning		
			DSC ↑	IoU ↑	Assd ↓	DSC ↑	IoU ↑	Assd ↓	DSC ↑	IoU ↑	Assd ↓	DSC ↑	IoU ↑	Assd ↓
SAMed (Zhang and Liu, 2023)			31.73	19.42	<u>40.08</u>	38.53	25.62	38.71	48.11	32.8	<u>31.24</u>	49.45	34.43	30.67
SAM-LST (Chai et al., 2024)			28.75	18.38	46.53	39.69	26.13	37.95	45.25	29.72	33.18	47.27	31.57	31.39
CaseNet (Yu et al., 2017)			<u>33.94</u>	<u>21.33</u>	45.02	39.05	25.36	<u>37.68</u>	46.83	31.27	34.29	49.87	33.81	31.09
DeepLabv3+ (Chen et al., 2018)			27.62	16.42	50.77	36.79	23.8	42.79	48.65	33.28	31.96	50.27	35.16	26.47
D ² GPLand+ (Ours)			42.15	27.4	37.12	45.55	30.53	34.26	53.96	37.72	29.06	55.94	39.4	23.65

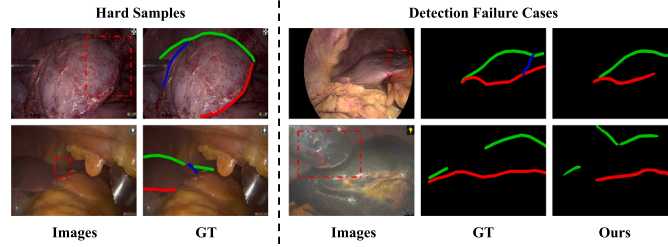


Fig. 13. Left: hard samples selected from L3D-2K. Right: failure landmark detection cases of D²GPLand+.

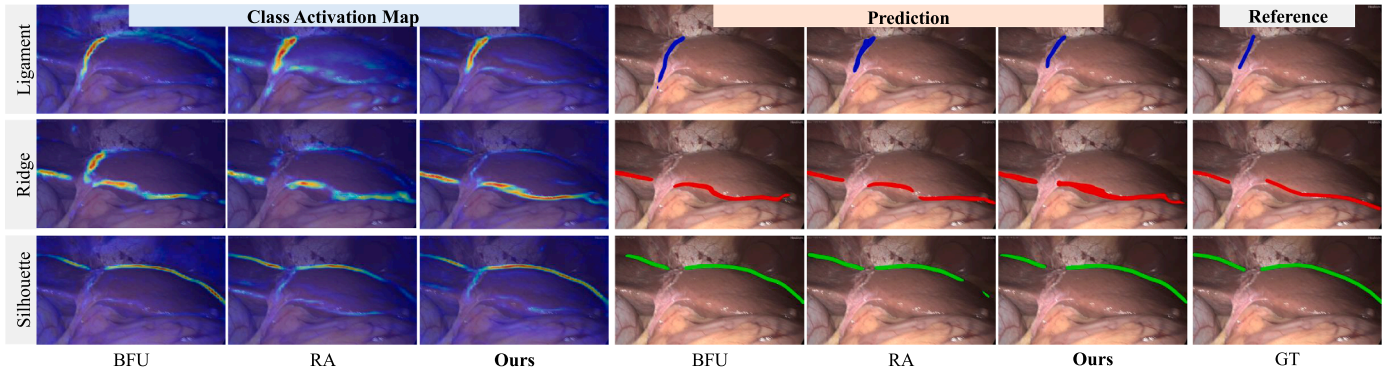


Fig. 14. Comparing class-aware activation maps of D²GPLand+ and D²GPLand (Pei et al., 2024). BFU denotes replacing CUMamba with the Bi-modal Feature Unification module and RA denotes replacing AFA module with the reverse attention.

dimensions for effective RGB-D fusion. Compared to the Bi-modal Feature Unification (BFU) module D²GPLand (Pei et al., 2024), CUMamba enhances long-range learning while retaining linear-time computational complexity by incorporating the State Space Model for feature fusion. The dual-scan strategy enables more comprehensive cross-modal feature fusion through aggregating informative cues on both spatial and channel dimensions. CUMamba not only preserves global geometric consistency but also sharpens local anatomical cues, which is better aligned with the demands of landmark detection. The comparisons in Section 5.3.3

further substantiate these advancements, showing consistent improvements of CUMamba over BFU.

We also introduce AFA module that embraces learnable granularity adjustment together with Laplacian filtering to strengthen the perception of ambiguous anatomical structures in implicit and explicit fashions. AFA adapts implicit frequency-domain enhancement and explicit Laplacian anatomical filtering to more comprehensively capture boundary features in both spatial and frequency domains. Besides, AFA integrates the learnable granularity adjustment to control the detail level

the features from both pathways with a group of lightweight learnable tokens, enabling the highlight of landmark-relevant anatomical features while suppressing the other boundary features. By jointly assembling these components, the AFA module can effectively sharpen anatomical clues and further strengthen the perception of related anatomical features. Experimental results further prove the effectiveness of the AFA module and its significant improvements (see Section 5.3.2). Our framework outperforms other competitors in both qualitative and quantitative evaluations (refer to Tables 2, 3 and Fig. 8), particularly in distinguishing ambiguous regions and preserving topological structures.

To illustrate the benefits of CUMamba and AFA, we apply the Grad-CAM (Selvaraju et al., 2017) to visualize the class-aware activation maps for three landmark categories in Fig. 14. For the two variants, CUMamba and AFA in D²GPLand+ are replaced with BFU and RA, respectively. When using BFU, the absence of global modeling causes the attention to be misled onto surrounding textures and off-structure emphasis, such as scattered hot spots on the background in the attention map for the ridge. With RA, the model struggles to preserve topological context, leading to discontinuities in the ridge and silhouette maps. In contrast, D²GPLand+ yields compact and spatially coherent attention that closely aligns with landmark boundaries while effectively suppressing outliers. This produces more complete topological relationships and enhances perceptual reliability in ambiguous target regions.

Although our method achieves superior performance, there is still space for improvement. On one fold, the performance in detecting small landmark regions can be further enhanced. As shown in the top sample of Fig. 13 (right), when the camera is far away from the liver, the model fails to accurately identify the ligament and a portion of the ridge. In future upgrades, we strive to address this issue by incorporating multi-scale fine-grained feature perception into the framework. In the bottom sample, our model struggles to discriminate some atypical representations, such as misidentifying surgical instruments as part of the silhouette and missing a part of the ridge due to abnormal liver coloration. A potential solution is to distill semantics from large vision-language models for promoting representation learning. Considering the goal of real-time intraoperative assistance, future work will continue to improve detection accuracy while exploring lightweight designs to increase the feasibility of clinical deployment.

7. Conclusion

In this paper, we propose a depth-induced prompt learning framework D²GPLand+ for laparoscopic liver landmark detection. By incorporating depth-aware prompt embeddings, cross-dimension unified Mamba, and anatomical feature augmentation, our approach captures intrinsic geometric cues and strengthens RGB-D cross-modal perception and integration. Additionally, we introduce a novel liver landmark detection dataset L3D-2K, to advance intraoperative assisted research. Experimental results demonstrate the effectiveness of our method in capturing anatomical information for liver landmark detection. We hope that our work will contribute to preoperative-to-intraoperative tissue fusion, providing surgeons with intuitive guidance during laparoscopic procedures.

CRedit authorship contribution statement

Ruize Cui: Writing – original draft, Visualization, Methodology, Investigation; **Weixin Si:** Validation, Resources, Project administration, Data curation; **Zhixi Li:** Validation, Formal analysis, Data curation; **Kai Wang:** Validation, Resources, Investigation, Data curation; **Jialun Pei:** Writing – review & editing, Validation, Methodology; **Pheng-Ann Heng:** Writing – review & editing, Supervision, Resources, Investigation, Funding acquisition; **Jing Qin:** Writing – review & editing, Supervision, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The Co-author, Pheng Ann Heng, has previously collaborated with AEs of this journal, including Qi Dou, Stefanie Speidel, and Nassir Navab. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Adagolodjo, Y., Trivisonne, R., Haouchine, N., Cotin, S., Courtecuisse, H., 2017. Silhouette-based pose estimation for deformable organs application to surgical augmented reality. In: IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 539–544.
- Ali, S., Espinel, Y., Jin, Y., Liu, P., Güttner, B., Zhang, X., Zhang, L., Dowrick, T., Clarkson, M.J., Xiao, S., et al., 2025. An objective comparison of methods for augmented reality in laparoscopic liver resection by preoperative-to-intraoperative image fusion from the international conference on medical image computing and computer-assisted intervention 2022 challenge. *Med. Image Anal.* 99, 103371.
- Borgars, J., Raja, J., Ramakrishnan, A., Abbas, A.K., Gallagher, A., Mohamad Shahir, A.N., Vraimakis, T., Ali, S., 2025. Intraoperative segmentation through deep learning and mask post-processing in laparoscopic liver surgery. In: Annual Conference on Medical Image Understanding and Analysis, pp. 204–218.
- Cao, H., Wang, Y., Chen, J., Jiang, D., Zhang, X., Tian, Q., Wang, M., 2022. Swin-UNet: UNet-like pure transformer for medical image segmentation. In: European Conference on Computer Vision.
- Chai, S., Jain, R.K., Teng, S., Liu, J., Li, Y., Tateyama, T., Chen, Y.-w., 2024. Ladder fine-tuning approach for sam integrating complementary network. *Procedia Comput. Sci.* 246, 4951–4958.
- Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A.L., Zhou, Y., 2021. TransUNet: transformers make strong encoders for medical image segmentation. *arXiv preprint arXiv:2102.04306*.
- Chen, L.-C., Zhu, Y., Papandreou, G., Schroff, F., Adam, H., 2018. Encoder-decoder with atrous separable convolution for semantic image segmentation. In: European Conference on Computer Vision.
- Chen, T., Kornblith, S., Norouzi, M., Hinton, G., 2020. A simple framework for contrastive learning of visual representations. In: International Conference on Machine Learning, pp. 1597–1607.
- Chen, T., Zhu, L., Deng, C., Cao, R., Wang, Y., Zhang, S., Li, Z., Sun, L., Zang, Y., Mao, P., 2023. Sam-adapt: adapting segment anything in underperformed scenes. In: International Conference of Computer Vision, pp. 3367–3375.
- Chen, Y., Li, Z., Xu, C., Liu, A.Q., Cui, R., Xu, X., Teoh, J. Y.-C., He, S., Qin, J., 2025. ProstaTD: bridging surgical triplet from classification to fully supervised detection. *arXiv preprint arXiv:2506.01130*.
- Collins, T., Pizarro, D., Gasparini, S., Bourdel, N., Chauvet, P., Canis, M., Calvet, L., Bartoli, A., 2020. Augmented reality guided laparoscopic surgery of the uterus. *IEEE Trans. Med. Imaging* 40 (1), 371–380.
- Cui, R., Zhang, J., Pei, J., Wang, K., Heng, P.-A., Qin, J., 2025. Topology-constrained learning for efficient laparoscopic liver landmark detection. In: International Conference on Medical Image Computing and Computer-Assisted Intervention. Springer, pp. 585–594.
- Gu, A., Dao, T., 2023. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*.
- Guo, D., Si, W., Li, Z., Pei, J., Heng, P.-A., 2025. Surgical workflow recognition and blocking effectiveness detection in laparoscopic liver resection with pringle maneuver. In: Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 39, pp. 3220–3228.
- Han, J.J., Acar, A., Henry, C., Wu, J.Y., 2024. Depth anything in medical images: A comparative study. *arXiv preprint arXiv:2401.16600*.
- Han, Z., Zhou, J., Pei, J., Qin, J., Fan, Y., Dou, Q., 2025. Towards reliable AR-guided surgical navigation: interactive deformation modeling with data-driven biomechanics and prompts. *IEEE Trans. Med. Imaging* 44 (11), 4417–4431.
- He, K., Zhang, X., Ren, S., Sun, J., 2015. Deep residual learning for image recognition. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition.
- He, X., Cao, K., Zhang, J., Yan, K., Wang, Y., Li, R., Xie, C., Hong, D., Zhou, M., 2025. Pan-Mamba: effective pan-sharpening with state space model. *Inf. Fusion* 115, 102779.
- Hu, X., Xu, X., Shi, Y., 2023. How to efficiently adapt large segmentation model (SAM) to medical images. *arXiv preprint arXiv:2306.13731*.
- Huang, H., Lin, L., Tong, R., Hu, H., Zhang, Q., Iwamoto, Y., Han, X., Chen, Y.-W., Wu, J., 2020. UNet 3+: a full-scale connected unet for medical image segmentation. In: ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 1055–1059.

- Isensee, F., Wald, T., Ulrich, C., Baumgartner, M., Roy, S., Maier-Hein, K., Jaeger, P.F., 2024. nnU-Net revisited: a call for rigorous validation in 3d medical image segmentation. In: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 488–498.
- Kasai, M., Uchiyama, H., Aihara, T., Ikuta, S., Yamanaka, N., 2023. Laparoscopic projection mapping of the liver portal segment, based on augmented reality combined with artificial intelligence, for laparoscopic anatomical liver resection. *Cureus* 15 (11), e48450.
- Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.-Y., Dollár, P., Girshick, R., 2023. Segment anything. In: International Conference of Computer Vision, pp. 4015–4026.
- Koo, B., Özgür, E., Le Roy, B., Buc, E., Bartoli, A., 2017. Deformable registration of a pre-operative 3D liver volume to a laparoscopy image using contour and shading cues. In: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 326–334.
- Koo, B., Robu, M.R., Allam, M., Pfeiffer, M., Thompson, S., Gurusamy, K., Davidson, B., Speidel, S., Hawkes, D., Stoyanov, D., et al., 2022. Automatic, global registration in laparoscopic liver surgery. *Int. J. Comput. Assist. Radiol. Surg.* 17, 167–176.
- Labrunie, M., Pizarro, D., Tilmant, C., Bartoli, A., 2023. Automatic 3D/2D deformable registration in minimally invasive liver resection using a mesh recovery network. In: Medical Imaging with Deep Learning.
- Labrunie, M., Ribeiro, M., Mourthadhoi, F., Tilmant, C., Le Roy, B., Buc, E., Bartoli, A., 2022. Automatic preoperative 3D model registration in laparoscopic liver resection. *Int. J. Comput. Assist. Radiol. Surg.* 17 (8), 1429–1436.
- Le Roy, B., Özgür, E., Koo, B., Buc, E., Bartoli, A., 2019. Augmented reality guidance in laparoscopic hepatectomy with deformable semi-automatic computed tomography alignment (with video). *J. Visc. Surg.* 156 (3), 261–262.
- Lee, T.S., 1996. Image representation using 2D gabor wavelets. *IEEE Trans. Pattern Anal. Mach. Intell.* 18 (10), 959–971.
- Li, H., Hu, Q., Yao, Y., Yang, K., Chen, P., 2024. CFMW: cross-modality fusion mamba for multispectral object detection under adverse weather conditions. *arXiv preprint arXiv:2404.16302*.
- Liu, H., Guo, L., Zhou, Z., Zhang, H., 2022. Pyramid-context guided feature fusion for RGB-D semantic segmentation. In: IEEE International Conference on Multimedia and Expo, pp. 1–6.
- Mou, L., Zhao, Y., Fu, H., Liu, Y., Cheng, J., Zheng, Y., Su, P., Yang, J., Chen, L., Frangi, A.F., et al., 2021. CS2-Net: deep learning segmentation of curvilinear structures in medical imaging. *Med. Image Anal.* 67, 101874.
- Özgür, E., Koo, B., Le Roy, B., Buc, E., Bartoli, A., 2018. Preoperative liver registration for augmented monocular laparoscopy using backward-forward biomechanical simulation. *Int. J. Comput. Assist. Radiol. Surg.* 13, 1629–1640.
- Park, S.-J., Hong, K.-S., Lee, S., 2017. RDFNet: RGB-D multi-level residual feature fusion for indoor semantic segmentation. In: International Conference of Computer Vision, pp. 4980–4989.
- Pei, J., Cui, R., Li, Y., Si, W., Qin, J., Heng, P.-A., 2024. Depth-driven geometric prompt learning for laparoscopic liver landmark detection. In: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 154–164.
- Pei, J., Zhang, J., Qin, G., Wang, K., Jin, Y., Heng, P.-A., 2025. Instrument-tissue-guided surgical action triplet detection via textual-temporal trail exploration. *IEEE Trans. Med. Imaging* 44 (12), 5278–5289.
- Plantefeve, R., Haouchine, N., Radoux, J.-P., Cotin, S., 2014. Automatic alignment of pre and intraoperative data using anatomical landmarks for augmented laparoscopic liver surgery. In: IEEE International Symposium on Biomedical Imaging, pp. 58–66.
- Ronneberger, O., Fischer, P., Brox, T., 2015. U-Net: convolutional networks for biomedical image segmentation. In: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 234–241.
- Schneider, C., Allam, M., Stoyanov, D., Hawkes, D.J., Gurusamy, K., Davidson, B.R., 2021. Performance of image guided navigation in laparoscopic liver surgery—a systematic review. *Surg. Oncol.* 38, 101637.
- Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D., 2017. Grad-CAM: visual explanations from deep networks via gradient-based localization. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 618–626.
- Smithmaitrie, P., Khaonualsri, M., Sae-Lim, W., Wangkulangkul, P., Jearanai, S., Cheewatanakornkul, S., 2024. Development of deep learning framework for anatomical landmark detection and guided dissection line during laparoscopic cholecystectomy. *Heliyon* 10 (3), e25210.
- Tziafas, G., Kasaei, H., 2023. Early or late fusion matters: efficient RGB-D fusion in vision transformers for 3D object recognition. In: IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 9558–9565.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Jones, A. N., Kaiser, Ł., Polosukhin, I., 2017. Attention is all you need. In: Advances in Neural Information Processing Systems 30.
- Wang, J., Sun, K., Cheng, T., Jiang, B., Deng, C., Zhao, Y., Liu, D., Mu, Y., Tan, M., Wang, X., et al., 2020. Deep high-resolution representation learning for visual recognition. *IEEE Trans. Pattern Anal. Mach. Intell.* 43 (10), 3349–3364.
- Wang, Y., Song, K., Liu, Y., Ma, S., Yan, Y., Carneiro, G., 2025. Leveraging labelled data knowledge: a cooperative rectification learning network for semi-supervised 3D medical image segmentation. *Med. Image Anal.* , 103461.
- Wu, J., Fu, R., Fang, H., Liu, Y., Wang, Z., Xu, Y., Jin, Y., Arbel, T., 2023. Medical sam adapter: Adapting segment anything model for medical image segmentation. *arXiv preprint arXiv:2304.12620*.
- Xiao, X., Lian, S., Luo, Z., Li, S., 2018. Weighted res-UNet for high-quality retina vessel segmentation. In: International Conference on Information Technology in Medicine and Education, pp. 327–331.
- Yang, L., Kang, B., Huang, Z., Zhao, Z., Xu, X., Feng, J., Zhao, H., 2024. Depth anything v2. *arXiv preprint arXiv:2406.09414*.
- Yin, W., Zhang, J., Wang, O., Niklaus, S., Chen, S., Liu, Y., Shen, C., 2022. Towards accurate reconstruction of 3D scene shape from a single monocular image. *IEEE Trans. Pattern Anal. Mach. Intell.* 45 (5), 6480–6494.
- Yu, Z., Feng, C., Liu, M.-Y., Ramalingam, S., 2017. CaseNet: deep category-aware semantic edge detection. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 5964–5973.
- Zhang, H., Bulathsinhala, S., Davidson, B.R., Clarkson, M.J., Ramalhinho, J., 2025a. Deep hashing for global registration of preoperative CT and video images for laparoscopic liver surgery. *Int. J. Comput. Assist. Radiol. Surg.* 20, 1461–1469.
- Zhang, K., Liu, D., 2023. Customized segment anything model for medical image segmentation. *arXiv preprint arXiv:2304.13785*.
- Zhang, T., Zheng, S., Cheng, J., Jia, X., Bartlett, J., Cheng, X., Qiu, Z., Fu, H., Liu, J., Leonardis, A., et al., 2024. Structure and intensity unbiased translation for 2D medical image segmentation. *IEEE Trans. Pattern Anal. Mach. Intell.* 46 (12), 10060–10075.
- Zhang, X., Ali, S., Kang, Y., Zhu, J., Han, M., Wang, L., Wang, X., Zhang, L., 2025b. Liver mask-guided SAM-enhanced dual-decoder network for landmark segmentation in AR-guided surgery. *Int. J. Comput. Assist. Radiol. Surg.* <https://doi.org/10.1007/s11548-025-03516-9>
- Zhou, J., Gao, B., Wang, K., Pei, J., Heng, P.-A., Qin, J., 2025. Landmark-free preoperative-to-intraoperative registration in laparoscopic liver resection. *IEEE Trans. Med. Imaging* 44 (11), 4350–4362.
- Zhou, Z., Siddiquee, M. M.R., Tajbakhsh, N., Liang, J., 2019. UNet++: redesigning skip connections to exploit multiscale features in image segmentation. *IEEE Trans. Med. Imaging* 39 (6), 1856–1867.
- Zhu, L., Liao, B., Zhang, Q., Wang, X., Liu, W., Wang, X., 2024. Vision Mamba: efficient visual representation learning with bidirectional state space model. In: Proceedings of the 41st International Conference on Machine Learning.